Exercise 2
SQL Aggregate x Operation

1 List distinct departments in the students table

```
SELECT DISTINCT(department)
FROM students
```

department
IT
HR
Finance

2 Get the average age of student per department

```
SELECT department, AVG(age) AS avg_age
FROM student
GROUP BY department
```

department	avg_age
IT	20.5
HR	22.0
Finance	23.0

3 Show departments with more than 1 student

```
SELECT department, COUNT(student_id) AS student_count
FROM students
GROUP BY department
HAVING COUNT(student_id) > 1;
```

department	Student_Count
IT	2
HR	2

4 Get all students whose age is between 21 and 23

```
SELECT Student_id, name, age, department
FROM Students
WHERE age BETWEEN 21 and 23;
```

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all students in the IT OR HR department Who are older than 21

```
SELECT Student_id, name, age, department
FROM Students
WHERE department IN ('IT', 'HR') AND age > 21;
```

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6 Show total Credits per department only for departments with more than 5 total Credits

```
SELECT department, Sum(Credits) as total_Credits
FROM Courses
GROUP BY department
HAVING Sum(Credits) > 5;
```

Course_id Course

department	total_credits
IT	11

7. list all Courses that do not have 4 Credits

```
SELECT Course_id, Course_name, department, Credits
FROM Courses
WHERE Credits != 4;
```

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
105	Statistics	HR	3
164	Exal	Finance	2

8 Show top 3 Courses by Credits in descending order

```
SELECT Course_id, Course_name, Credits
FROM Courses
ORDER BY Credits DESC
LIMIT 3;
```

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basic	3

9 Get the maximum, minimum and average grade across all enrollments

```
SELECT MAX(grade) AS max_grade, MIN(grade) AS min_grade, AVG(grade)
AS avg_grade
FROM enrollments
```

max_grade	min_grade	avg_grade
90	78	84.6

10 Count how many enrollment exist per Course

```
SELECT Course_id, COUNT(enrollments_id) AS enrollment_count
FROM enrollents
GROUP BY Course_id
```

Course_id	enrollment_Count
101	1
102	1
103	1
104	1
105	1

11 Find total Salary and total bonus per department

```
SELECT department, Sum(Salary) AS total_Salary, Sum(bonus) AS total_bonus
FROM Salaries
GROUP BY department
```

department	total_Salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12 Show departments where average Salary is above 55000

```
SELECT department, AVG(Salary) AS avg_Salary
FROM Salaries
GROUP BY department
HAVING AVG(Salary) > 55000;
```

13 list employees whose Salary plus bonus is greater than 60000

```
SELECT employee_id, name, Salary, bonus, (Salary + bonus) AS total_compensation
FROM Salaries
WHERE (Salary + bonus) > 60000;
```

employee_id	name	Salary	bonus	total_Compensation
1	Tom	60000	5000	65000
2	Spike	70000	6000	76000
3	Tyke	62000	5500	67500

14 Show total and average budget per department. Only include departments with an average budget above 70000

```
SELECT department, SUM(budget) AS total_budget, AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

15 List all projects with budgets between 50000 and 120000 excluding the marketing department

```
SELECT project_id, project_name, department, budget
FROM projects
```

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	HR Portal	HR	50000